Panskura Banamali College (Autonomous)

M.Sc. 1st Semester Examination 2022 (Online mode)

Subject : Computer Science Paper: COS-101

Paper Name : Data Structure and Algorithm

F.M. : 40 Time : 2 Hours

A. Answer any two questions: 2×4=8

1. **What is complexity of an algorithm? How do you push elements in a linked stack .**
2. **Define RPN? Write the POLISH algorithm.**
3. **What is the difference between tree & graph ?**

B. Answer any two questions: 2×8=16

1. **a) Write an algorithm which accepts a stack as input and reverse the element of the stack using one additional queue.**
   b) Prove that E=I+2q where E is the external path length , I is the internal path length and q is the number of vertices that are not leaves in a tree.
   4+4
2. **a) Construct a binary search tree from the given values .**
   Consider the 1st value as the root value.
   Values – 45,23,29,85,92,7,11,35,49,51
   b)Prove that a linked binary tree with n nodes ,n ≥0 has exactly (n+1) NULL links.
   4+4
3. **a) Difference between complete binary tree and almost complete binary tree.**
   b) What is two-way list? Write the advantages of two-way list over signal linked-list. 4+4

C. Answer any one question : 1×16=16

1. Write short note (any four)

- **a) Garbage collection**
- **b) Quadratic probing and double hashing**
- **c) Heeder node in linked list**
- **d) linear probing.**
- **e) Priority queue.**

2. Write an algorithm to implement bubble sort & sort the following list in descending order by using the algorithm-

15, 12, 4, 16, 9, 10, 94

***